

Item	Component	Key Specifications	Qty	Unit Price (₹)	Total (₹)
1	ESP32 DEVKIT V1	Dual-core 240MHz, Wi-Fi 802.11 b/g/n, Bluetooth v4.2/BLE, 3.3V Logic.	1	400	400
2	ESP32-CAM	OV2640 2MP Camera, built-in microSD card slot, Wi-Fi/BT, internal flash.	1	450	450
3	TB6612FNG Motor Driver	Dual-channel, 1.2A continuous (3.2A peak) per channel, VCC up to 15V.	1	200	200
4	12V DC Gear Motor	~100-300 RPM, Metal gearbox, high torque (ideal for outdoor rovers).	2	250	500
5	NEO-6M GPS Module	UART interface (9600 baud), ceramic patch antenna, 3.3V/5V compatible.	1	500	500
6	IR Obstacle Sensor	LM393 comparator, adjustable digital output (High/Low), 2-30cm range.	2	50	100
7	Solar Panel	6V, 2W/3W Polycrystalline, bare wire or barrel jack output.	1	200	200
8	18650 Li-Ion Battery	3.7V nominal, 2500mAh+ capacity, non-protected (let BMS handle protection).	3	200	600
9	18650 Battery Holder	3-cell configuration. Must be wired in SERIES to output 11.1V for motors.	1	60	60
10	TP4056 Charger	5V 1A Micro/Type-C input, 4.2V cut-off. (See warning below).	1	40	40
11	LM2596 Buck Module	Input 4V-35V, Output 1.23V-30V adjustable, 3A max. Used to step 12V down to 5V.	1	100	100
12	0.96" OLED Display	128x64 pixels, SSD1306 driver, I2C interface (SDA/SCL), 3.3V/5V logic.	1	220	220
				Total	~₹ 3,370